

Cara

AI-Powered Assistant for Cosmetic Skin Analysis

POC Presentation

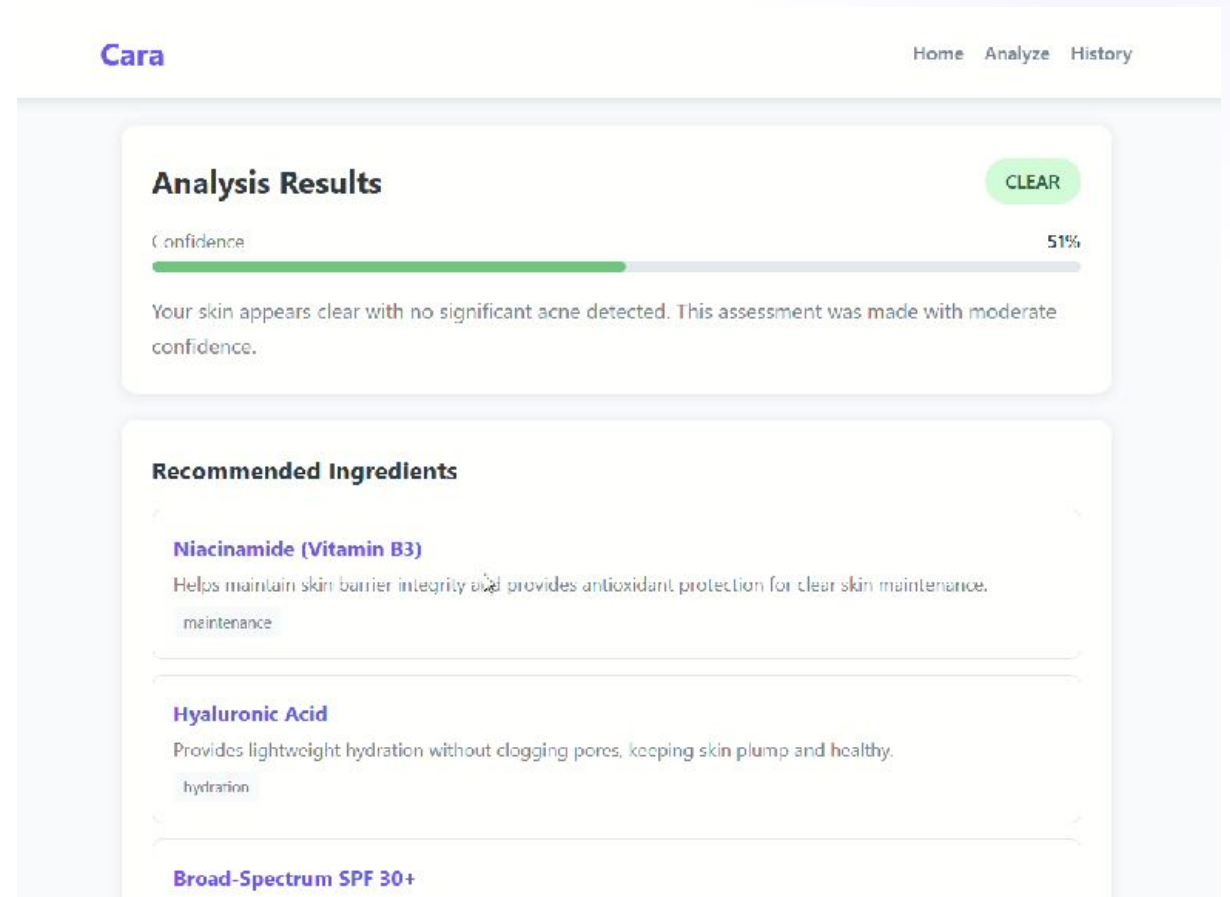
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MISSION & VISION

Explainable cosmetic skin analysis through modular architecture and advanced computer vision.

- ✓ **Production-Ready POC:** Validating E2E AI pipeline.
- ✓ **Modular ML:** Separated inference and logic layers.
- ✓ **Explainability:** Actionable skincare insights, not medical advice.



SPRINT 1-2: ARCHITECTURE RESEARCH



Each member explored a distinct architecture for 2 sprints to determine the optimal balance of inference speed, pre-training advantages, and classification accuracy on the ACNE04 dataset.

MODEL PERFORMANCE BENCHMARKING



Conclusion: While ViT proved to be a very strong architecture taking second place, EfficientNet ultimately won. Because EfficientNet is pre-trained on a massive amount of parameters, its initial weights were already incredibly sharp, allowing it to easily outperform the others with minimal training on our specific dataset size.

WHY EFFICIENTNET?



Pre-Trained Power: Leveraged massive pre-trained parameters, providing highly optimized, sharp weights out-of-the-box.



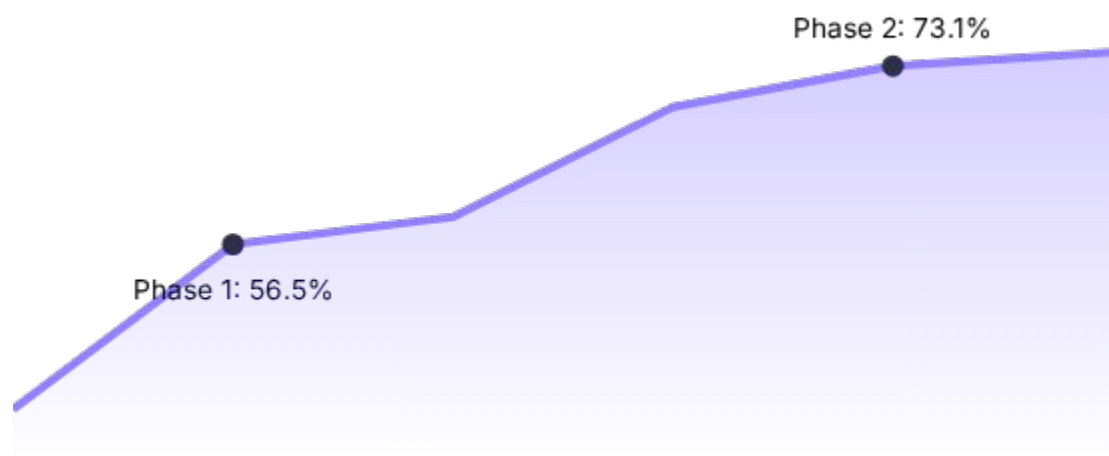
Fast Convergence: Beat competing models with very little training, requiring only minor fine-tuning.



Superior Scaling: Compound scaling methodology perfectly suited our dataset constraints.



TRAINING PROGRESSION: DEEP DIVE



Two-Phase Training Strategy

Phase 1: Frozen Backbone

10 Epochs leveraging sharp pre-trained weights. Val Acc: 56.5%.

Phase 2: Fine-Tuning

5 Epochs unfreezing upper layers. Significant jump to 73.1% Val Acc.

Final Evaluation

Generalization on unseen test set: 69.5% accuracy.

MODULAR SYSTEM ARCHITECTURE



Frontend

Client application. Captured images sent to Backend Gateway via REST API.



Backend Gateway

Orchestrates the flow.
Coordinates between Model Service and Business Logic Layer.



Model Service

EfficientNet inference engine with specialized skincare preprocessing.



BLP Engine

Rule-based processing to translate severity scores into ingredient recommendations.



Report Gen

Dynamic generation of explainable reports with ingredients and reasoning.



Data Pipeline

Ensuring smooth handoffs between Deep Learning outputs and Full Stack rendering.

THE PIVOT: RESILIENCE & TEAMWORK

"Adapting our workflow to push forward together."

Due to the war and severe shifts in our availability, we realized the planned Git splits and isolated development wouldn't be realistic.

Because the core of this project is Deep Learning—not complex full-stack separation—we pivoted to **gathering online via Zoom to build the implementation together.**

We built the project, committed everything to Git as a unit, and successfully created the demo you see today.



END-TO-END IMPLEMENTATION FLOW



Capture



Preprocess



Inference



BLP Rules



Report

Business Logic Processing (BLP)

Translates raw model output (e.g., Severity: *Moderate*) into actionable ingredient recommendations.

Explainable Recommendations

Every result includes an educational note explaining the 'Why' behind the suggested ingredient.

CONCLUSION & SUCCESS METRICS

POC Achievements:

- Successfully built E2E working demo despite external challenges.
- Validated EfficientNet as the current superior backbone due to sharp pre-trained weights.
- Integrated rule-based BLP for explainability.

Looking Forward:

The architecture is set up perfectly for future scaling. By abstracting the Deep Learning models from the frontend logic, future teams can easily swap in new models or add specific condition classifiers without breaking the application stack.



Questions?

Thank you for your attention.

Cara: Explainable Skincare AI

POC Presentation Demo

